Properties of Matter

Aim: to observe and compare the properties of the three states of matter.

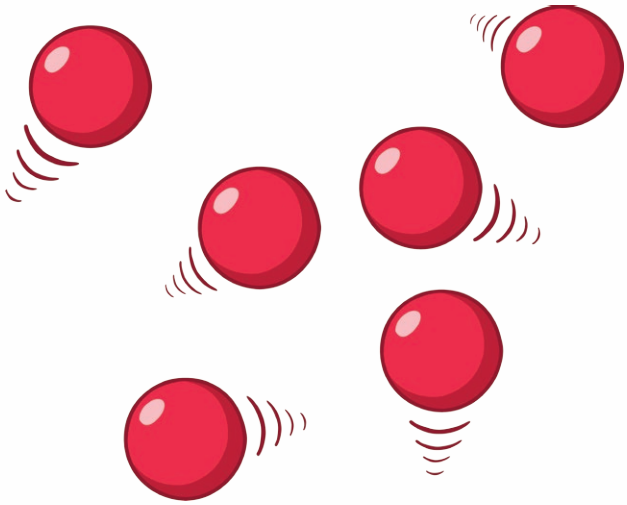
Equipment:

- Balloon filled with air
- Balloon filled with water
- Balloon filled with (set) plaster

Safety Information

Latex balloons – possible allergen.

Check with students and offer an alternative



Method:

Step 1: Taking care not to burst any of the balloons, try to twist or bend each one. Can you change its shape? Record your observations.

Step 2: With one hand, hold the balloon where it has been tied and gently push the contents away with the other. Can you compress the contents? (Again, take care not to burst any of the balloons.) Record your observations.

Step 3: Hold a balloon in each hand and compare their densities. Then swap one balloon and compare again. Keep comparing until you have decided which balloons are dense and which are not. Record your observations.

Step 4: Holding the balloon where it is tied, hold it just above the desk, then let go. Do the contents move and flow inside the balloon at all? Record your observations.

Step 5: Compare the balloons in any other ways you can think of and record your observations in the blank rows on the table.

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Result table

	Solid	Liquid	Gas
Can it change shape?			
Can it be compressed?			
It is dense (heavy)			
Does it flow?			

Questions

1. Describe each state of matter in a short paragraph, using the observations you have recorded in your table.
2. Draw the particle arrangement for each state of matter: solid, liquid and gas.
3. Using your particle diagram, explain why:
    a. A gas can be compressed
    b. A liquid can flow
    c. A solid is dense

Extra Challenge

Mix a tablespoon of cornflour with a few drops of water in a beaker. Stir until you have a smooth paste. What do you notice about the mixture when you stir and stop stirring? Does the mixture have properties like a solid, liquid or a gas? How would you classify this substance?

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Answers

Result table

	Solid	Liquid	Gas
Can it change shape?	**NO**	**YES**	**YES**
Can it be compressed?	**NO**	**NO**	**YES**
It is dense (heavy)	**YES**	**IN BETWEEN**	**NO**
Does it flow?	**NO**	**YES**	**YES**

Questions

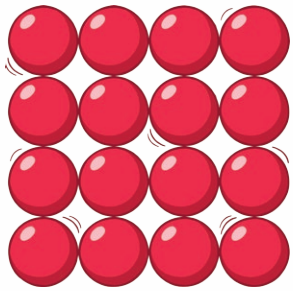
1. Describe each state of matter in a short paragraph, using the observations you have recorded in your table.

   A solid object has a fixed shape and volume, it cannot be compressed and does not flow. Solid objects are usually a high density.

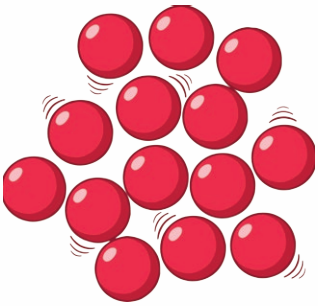
   A liquid object has a fixed volume but can change shape as it flows. It cannot be compressed and is usually a medium density.

   A gas object does not have a fixed shape or volume. It can flow and be compressed easily. Gas objects are not very dense.

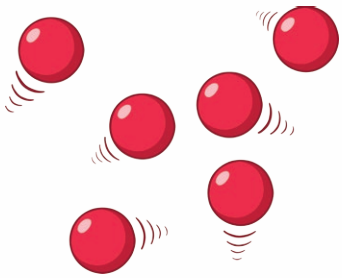
2. Draw the particle arrangement for each state of matter: solid, liquid and gas.



SOLID



LIQUID



GAS

3. Using your particle diagram, explain why:

   a. A gas can be compressed

As a gas is compressed, the particles move closer together and the spaces between particles becomes smaller.

b. A liquid can flow

The particles are arranged closely together, but not tightly packed and so they can move and slide past one another.

c. A solid is dense

Since the particles are arranged so close together, with very little space in between, there is a lot of mass in the volume.

Extra Challenge

Mix a tablespoon of cornflour with a few drops of water in a beaker. Stir until you have a smooth paste. What do you notice about the mixture when you stir and stop stirring? Does the mixture have properties like a solid, liquid or a gas? How would you classify this substance?

The cornflour, when mixed with water, forms a non-Newtonian substance and can exhibit properties of two states, depending on the pressure and force applied.

When the mixture is under pressure and being stirred, it exhibits properties of a solid, becoming hard and holding its shape.

When the mixture is not being stirred it exhibits properties of a liquid, becoming 'runny' and flowing (becoming less viscous).